

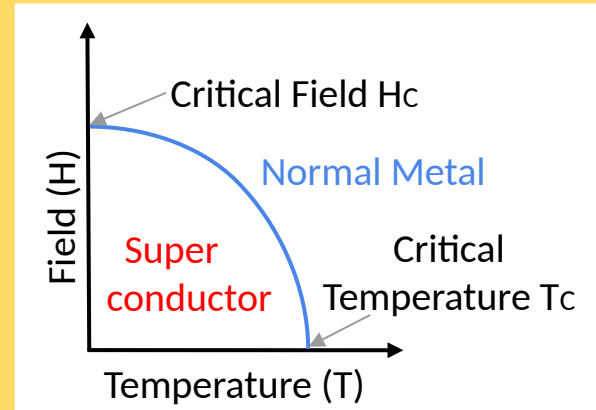
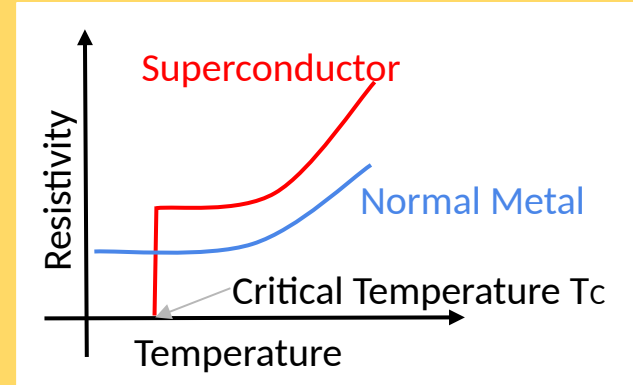
SUPERCONDUCTIVITY.

Define The Following:

24 times for 2 marks

- **Superconductivity** : Phenomenon of sudden disappearance of electrical resistance and expulsion of magnetic flux occurring in certain materials when they are cooled below a certain temperature.
- **Critical temperature** : The temperature below which the electrical resistance of a material suddenly drops to zero.
- **Critical Field** : The minimum field value at which normal resistivity is regained by the material and loses its superconducting state.

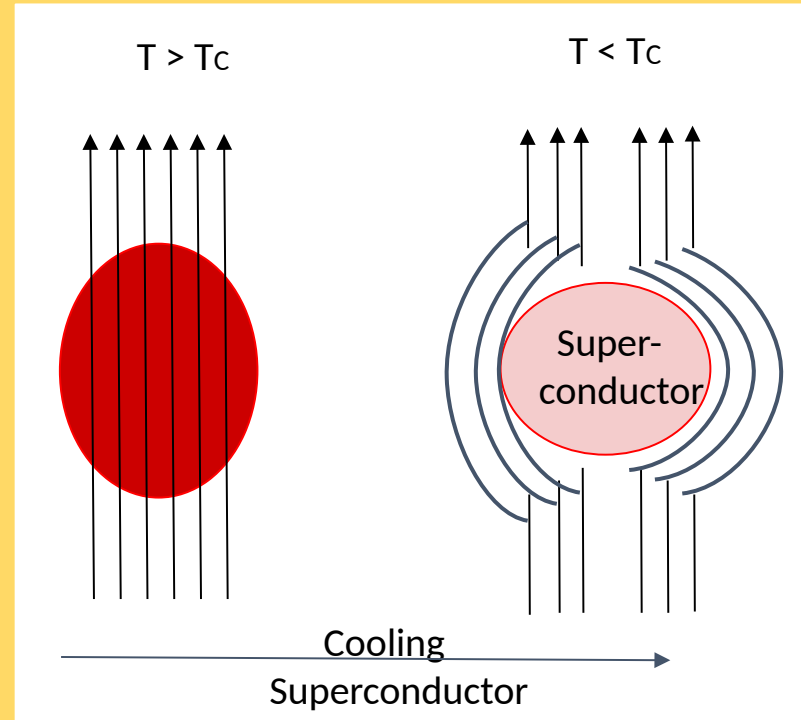
$$H_c = H_0[1-(T/T_c)^2]$$



Meissner Effect

- Superconducting material is cooled in a weak magnetic field ($H < H_c$), at critical temperature (T_c).
- Superconducting state: Field flux is expelled from the body of the material.
- This is because an equal and opposite magnetization is induced inside the body which throws out opposing field.
- Normal state: External magnetic field lines are able to penetrate through its body.
- The magnetic induction inside a material is given by: $B = \mu_0 (H + M)$

6 times for 5 marks



Applications of Meissner's Effect

- **Confirming Superconductivity:** When a material being cooled in a weak magnetic field demonstrates flux expulsion, it is a confirmed superconductor.
- **Levitated Frictionless Bearings:** The machines with rotating shafts ball bearings are covered with lubricants to reduce friction and thereby increase rotations per minute.

Perfect Diamagnets

- Susceptibility ($X=M/H$) is the magnetization per unit magnetic field.
- X for diamagnetic materials is negative as they get magnetized in the direction opposite to that of magnetic field.
- For superconductor to be in superconducting state, i.e. when $T < T_c$:

- $B=0$

- $H+M=0$

- $M=-H$.

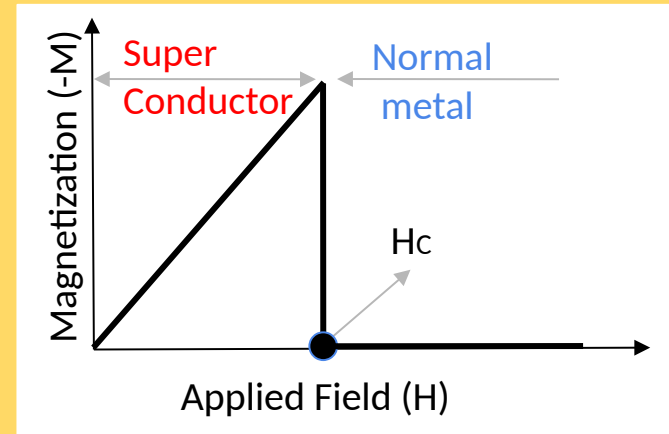
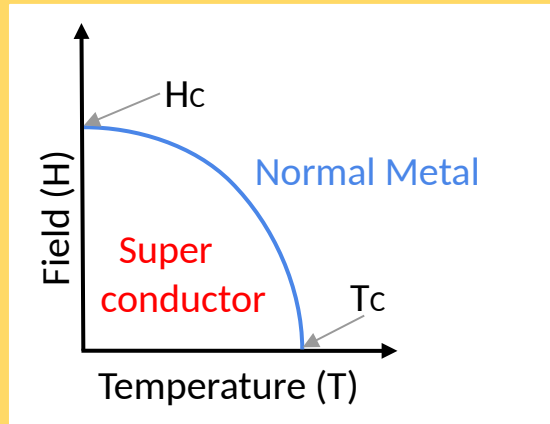
- $X = M/H = -1$

- As X is negative, it proves a superconductor is diamagnetic.

Type 1 Superconductors

7 times for 5 marks

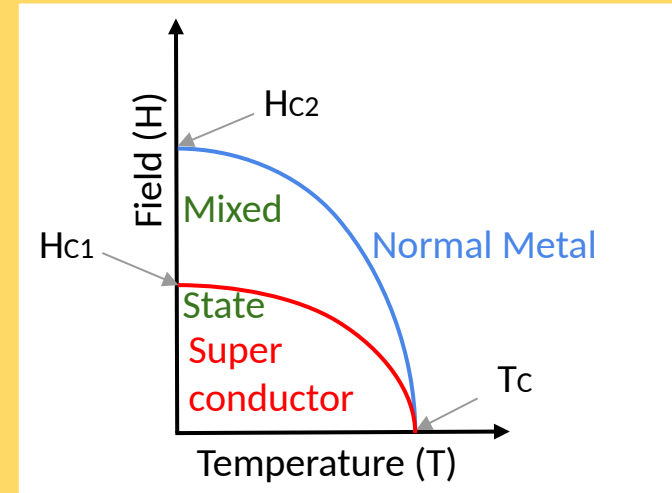
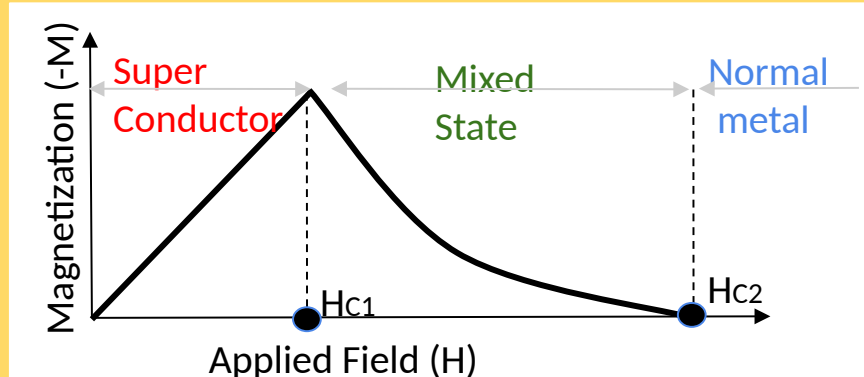
- Demonstrate perfect diamagnetic behavior in superconducting state.
- Are magnetized in direction opposite to the field.
- Obeys the Meissner effect.
- Low critical magnetic field.
- Exhibit single critical magnetic field.
- Also known as soft superconductors.



Type 2 Superconductors

6 times for 5 marks

- High critical temperature.
- High critical magnetic field.
- Partly obeys the Meissner effect.
- Exhibits two critical magnetic fields.
- Also known as hard superconductors.



SUPERCAPACITOR

Principle and Working

- Capacitor which has *high capacitance* and capable of changing and *storing energy at a higher density* than normal capacitor.

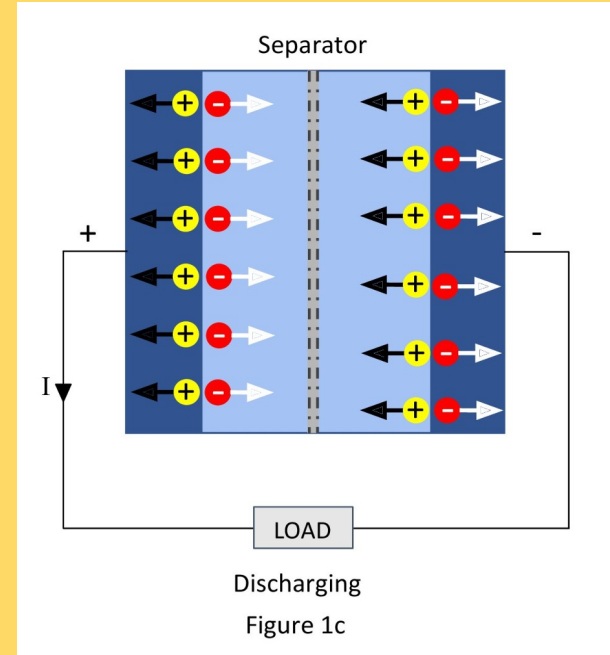
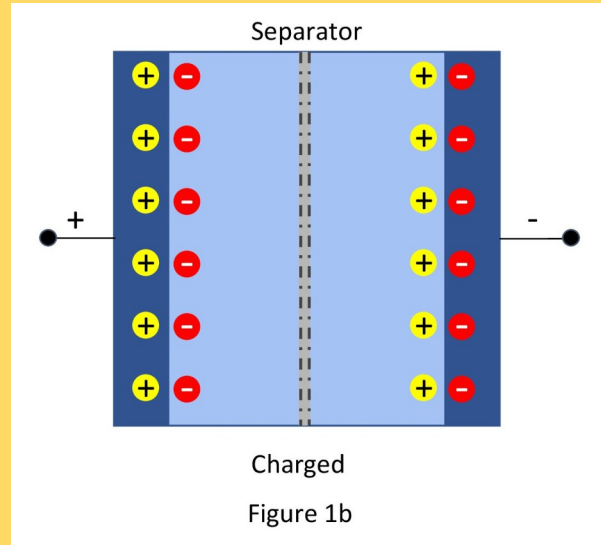
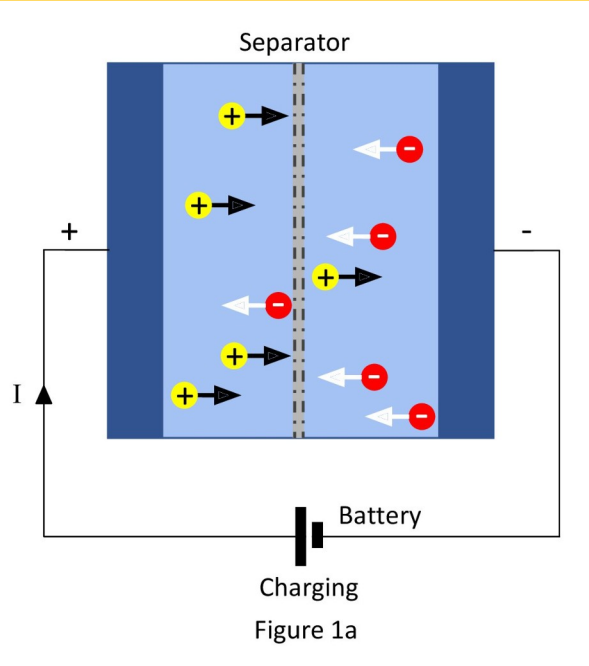
Principle:

- *Double layer capacitance* at the electrode - electrolyte interface principle, where electric charges are gathered on the electrode surface and ions of opposite charge are arranged on the electrolyte side.

Working:

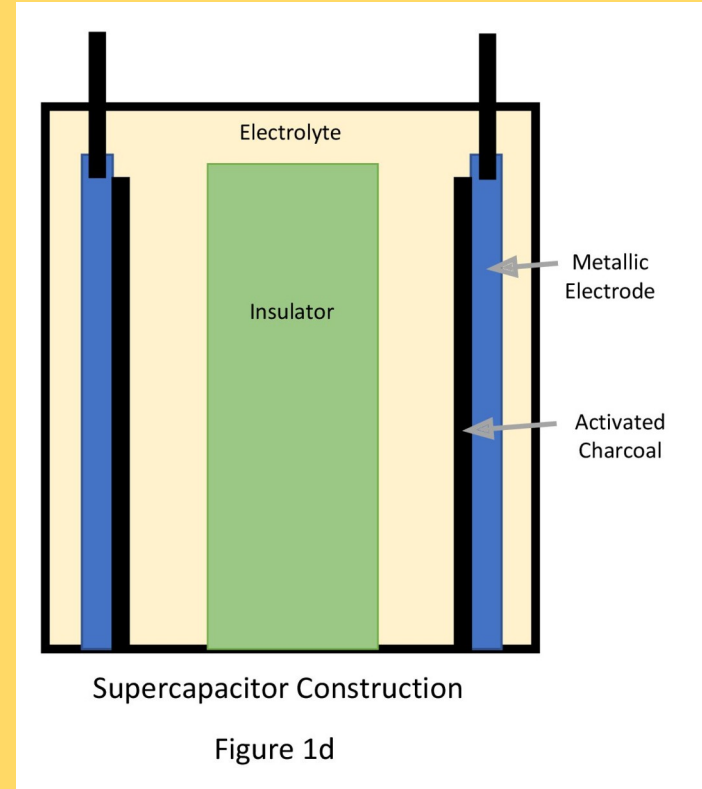
- The porous coating on the plates of a supercapacitor *stores a large amount of charge* because of the *added area*.
- *Opposite charges form on either side* of the separator, creating a double layer of charge.
- As the capacitance of a capacitor *increases* when the *distance between the plates decrease* and the *area of the plates increase*, a Supercapacitor achieves a much *higher capacitance* than any regular capacitor.

Working



Construction

- Plates are made of metals coated with a *porous substance*.
- Porous material gives *larger surface area*, increasing the storage capacity.
- Plates are soaked in an electrolyte and *separated* by a thin *insulator*.
- Post charging, layers of *opposite charges* are formed on *both sides of the separator*, thus forming the *double layer*.



Types of Supercapacitors

- *Double Layer Capacitors*: These capacitors achieve energy storage with *activated carbon electrodes*.
- *Pseudo-capacitors*: These capacitors store electrical energy by *electron charge transfer* between *electrodes and electrolyte*. Metal oxide or conductive polymer electrodes are used.
- *Hybrid Supercapacitor*: These have *two* electrodes; one stores charge *electrostatically* while other exhibits *electrochemical* capacitance.

Advantages and disadvantages

Advantages:

- Long-lasting
- Takes minimum amount of time to charge
- Not costly
- Fast release of battery
- Stores large amount of energy

Disadvantages:

- Low voltage for a single cell
- Energy density is low
- During discharge complete energy is not used
- Self - discharge is high
- Series connection of multiple capacitors is required for high voltage.

Applications

- Cellular Devices
- Uninterruptible Power Supplies (UPS)
- Industrial Lasers
- Medical Equipment
- Wireless communication Systems
- VCRs, CD Players
- Security systems, Computer Scanners
- Microwaves, Coffee Makers

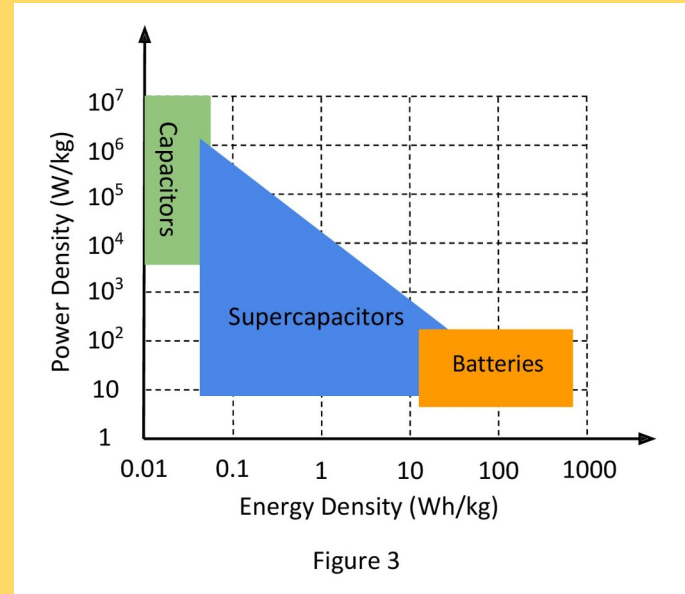
Energy and Power Density

Energy density:

- It is the amount of energy that can be stored per unit mass or per unit volume in any given device.
- A device with a high energy density value has the capacity to store a large amount of energy in a small mass or volume.

Power density:

- It is the amount of flow of energy per unit mass or volume, per unit time, in any given device.
- Tells us how quickly can the stored energy be utilised.



Batteries, Capacitors and Supercapacitors

Performance Parameters	Li-ion Battery	Ordinary Capacitor	Supercapacitor
Energy density (Wh/kg)	100-180 (HIGH)	0.01-0.02 (LOW)	4-10 (HIGH)
Power density (kW/kg)	300-3000 (LOW)	>200 (HIGH)	6000-60000 (HIGH)
Time to charge	15-60 mins	5-15s	2-10s
Lifespan	5-10 years	15-20 years	10 – 15 years